

CS4800: Designing Sustainable ICT Systems

Lecture 7.1: Sustainable Data Storage, Part 1

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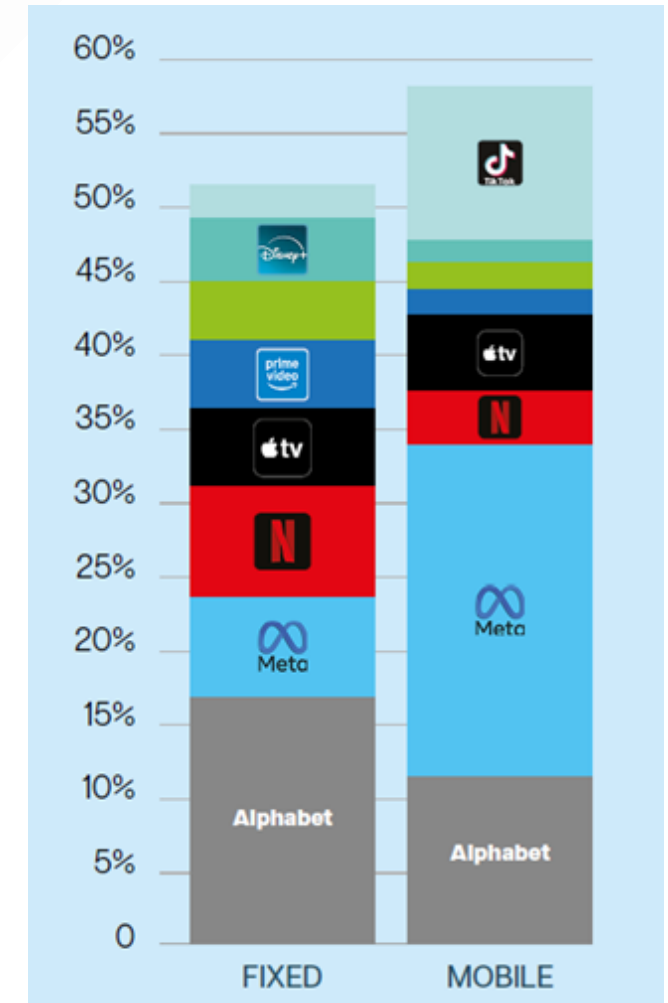
What will you learn in this lecture?

Lecture 7.1: Sustainable Data Storage, Part 1

- Drivers of data storage growth: consumer data, data retention, AI
- Data storage technologies: magnetic tape, optical discs, HDDs, SSDs
- Data storage architectures

Introduction: the datasphere

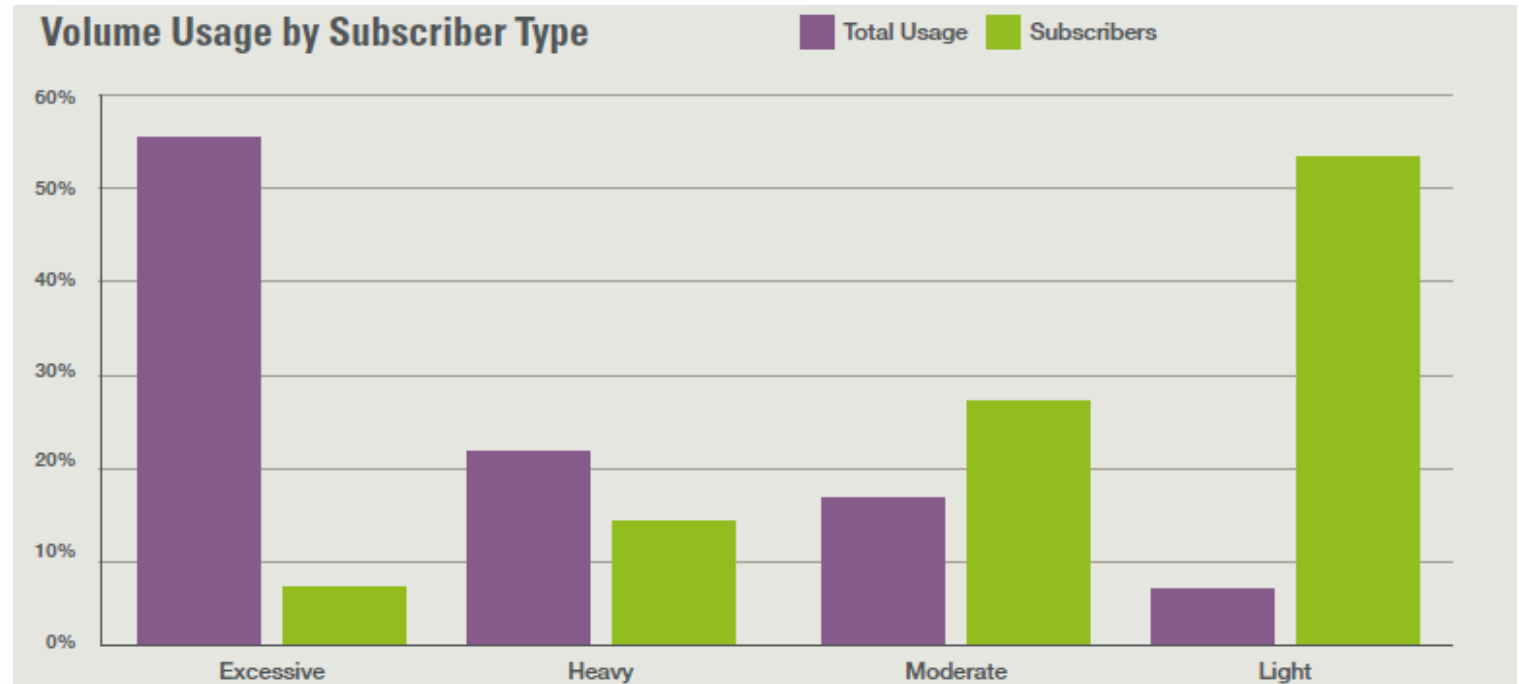
- 59 zettabytes (ZB: 8,000,000,000,000,000,000,000 bits) of data created in [2018](#)
- Various [estimates](#): hundreds of ZB to be created each year by 2030
- [Daily data volumes](#) (2022):
 - 5.9 million Google searches (3x versus 2013!)
 - 16 million text messages
 - 1 million hours of streaming video
 - 1.1 million swipes on Tinder



[Proportion of data volumes handled by Big Tech companies](#)

Introduction: the datasphere

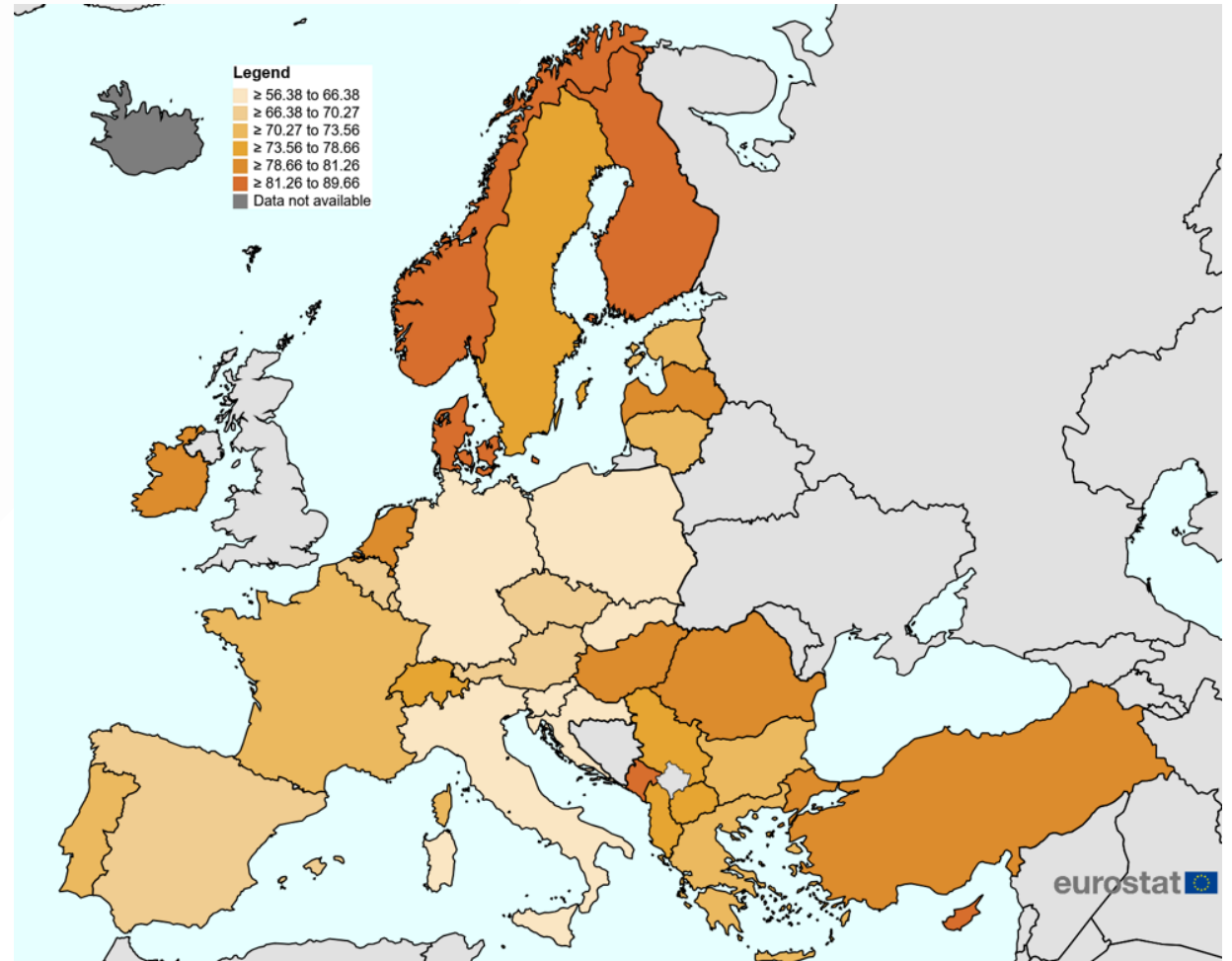
- Data creation and transmission:
 - Video streaming: YouTube, Netflix
 - Social media: Facebook, Instagram
 - Workspace: Google Drive, Zoom, Slack
- 23% year-on-year growth in traffic volume between 2021 and 2022
- 8% of users responsible for 55% of volume



Most data is used by a few power users

Drivers: consumer data

- US citizens tend to be [more online](#) than EU citizens:
 - 84% of US adults [use YouTube](#), 71% use Facebook
 - Some EU countries are [close in social media use](#) (Malta, 81.6%), while others not (Croatia, 61.5%)
 - Convergence among young people: 80% of US adults below 30 [use Instagram](#), while 89.3% of EU youths [use social media](#)



[Percentage of Europeans using social networks](#)

Drivers: consumer data

- What about the rest of the world?
- East Asia: local platforms dominant
 - China: WeChat (used by [87.3% of the population](#)), Douyin (78.4%)
 - Japan: Naver LINE (used by [85.3% of the population](#))
 - South Korea: Kakao Talk (47.9 million users, or [94% of smartphone users](#))



[East Asia features billions of social media users](#)

Drivers: data retention

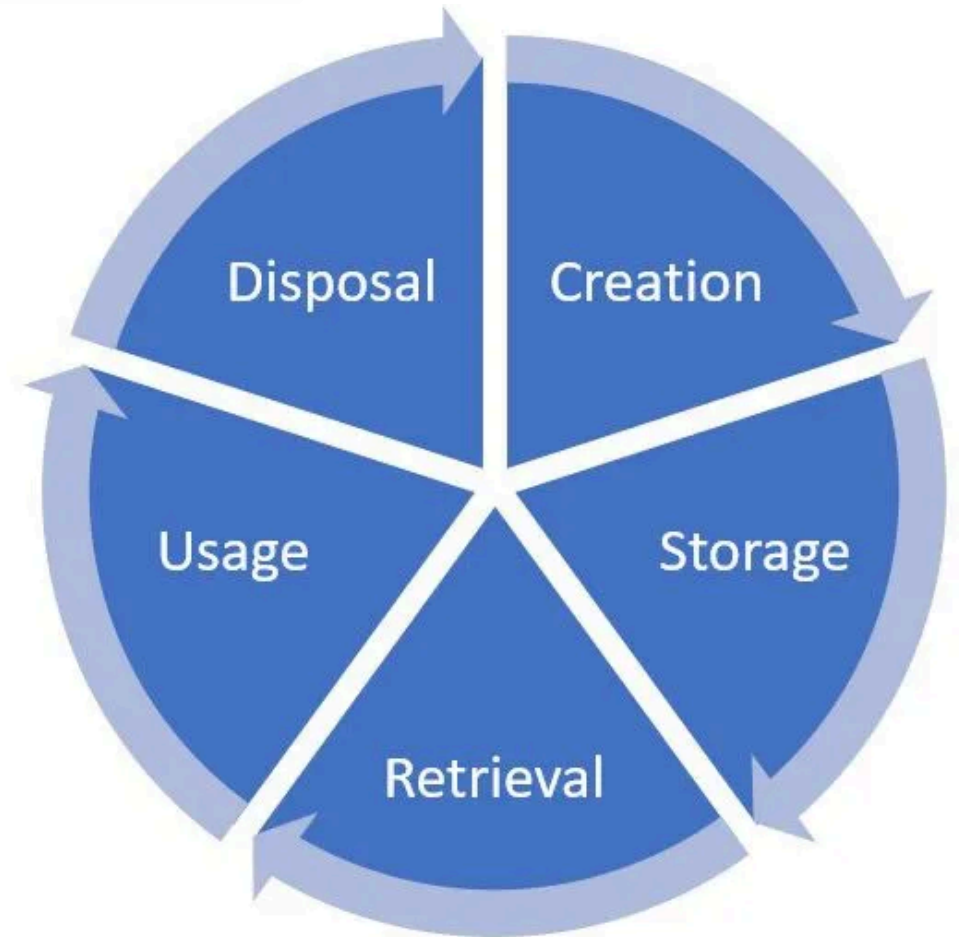
- Metadata storage by governments: complex [privacy](#) battleground
- Initial 2006 EU data retention directive [invalidated](#) by European Court of Justice
- Despite [privacy concerns](#), telecom providers still obliged to store communications metadata for years



Metadata can be just as useful as the content itself

Drivers: data retention

- Businesses legally obliged to store data:
 - Dutch [tax authority](#): store basic data (ledgers, accounts, payroll) for 7 years, data related to physical assets for 10 years
 - [GDPR](#): store personal data only for as long as required, and then verifiably destroy it
 - Business itself decides what 'as long as required' means!
- In total: enormous archiving capacity needed



Five steps in the data lifecycle

Drivers: AI training

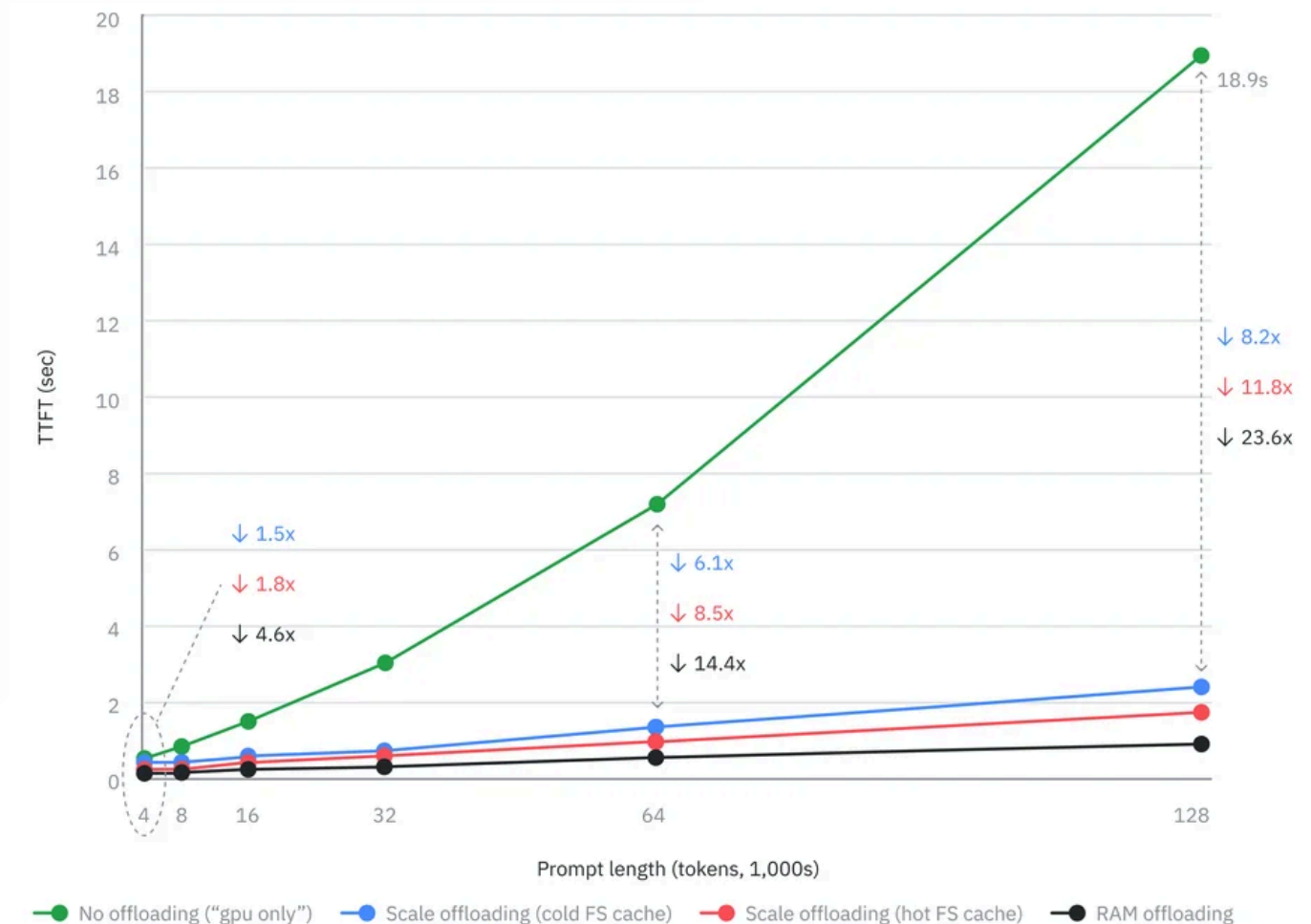
- Large quantities of data required for various AI training workloads
- Training [steps](#): data collection, cleaning, labeling, normalization, and tokenization
- Checkpointing: saving the state of the training job between expected failures
 - Up to [15 TB checkpoint size](#) for 1T parameter model!

Model	8B	70B	405B	1T
Data Parallelism	1	1	2	2
Total Processes	8	64	512	1024
Checkpoint size	105 GB	912 GB	5.29 TB	15 TB
Subset: 8-Process Size	105 GB	114 GB	94 GB	144 GB

[Training checkpoint size for various model sizes in the MLPerf Storage benchmark](#)

Drivers: AI inference

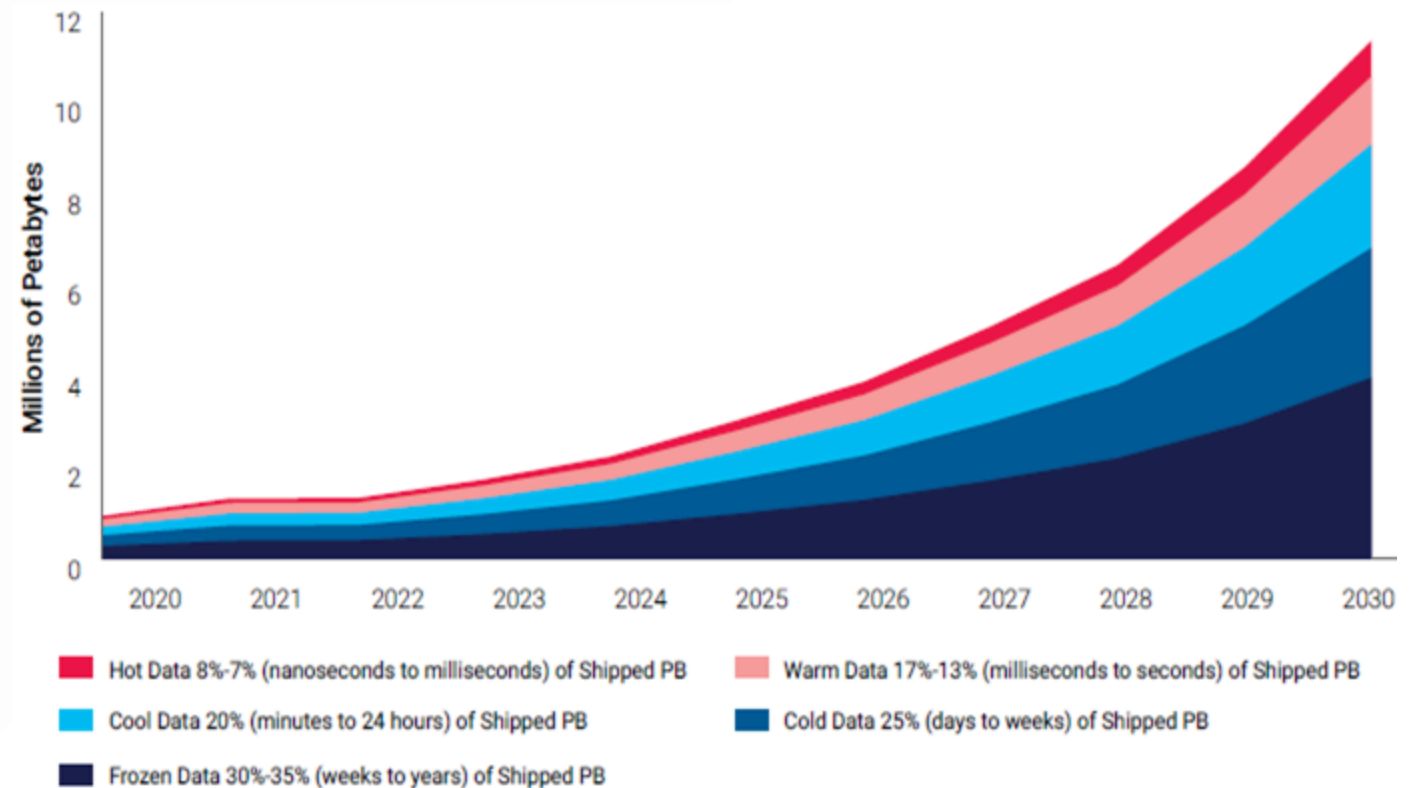
- LLM inference: large quantities of [intermediate runtime data](#)
- Increasing context size: insufficient CPU RAM and GPU memory → [offloading to local storage](#)
- High bandwidth and low latency cache access required



Processing time increases quadratically without GPU offloading.

Data storage

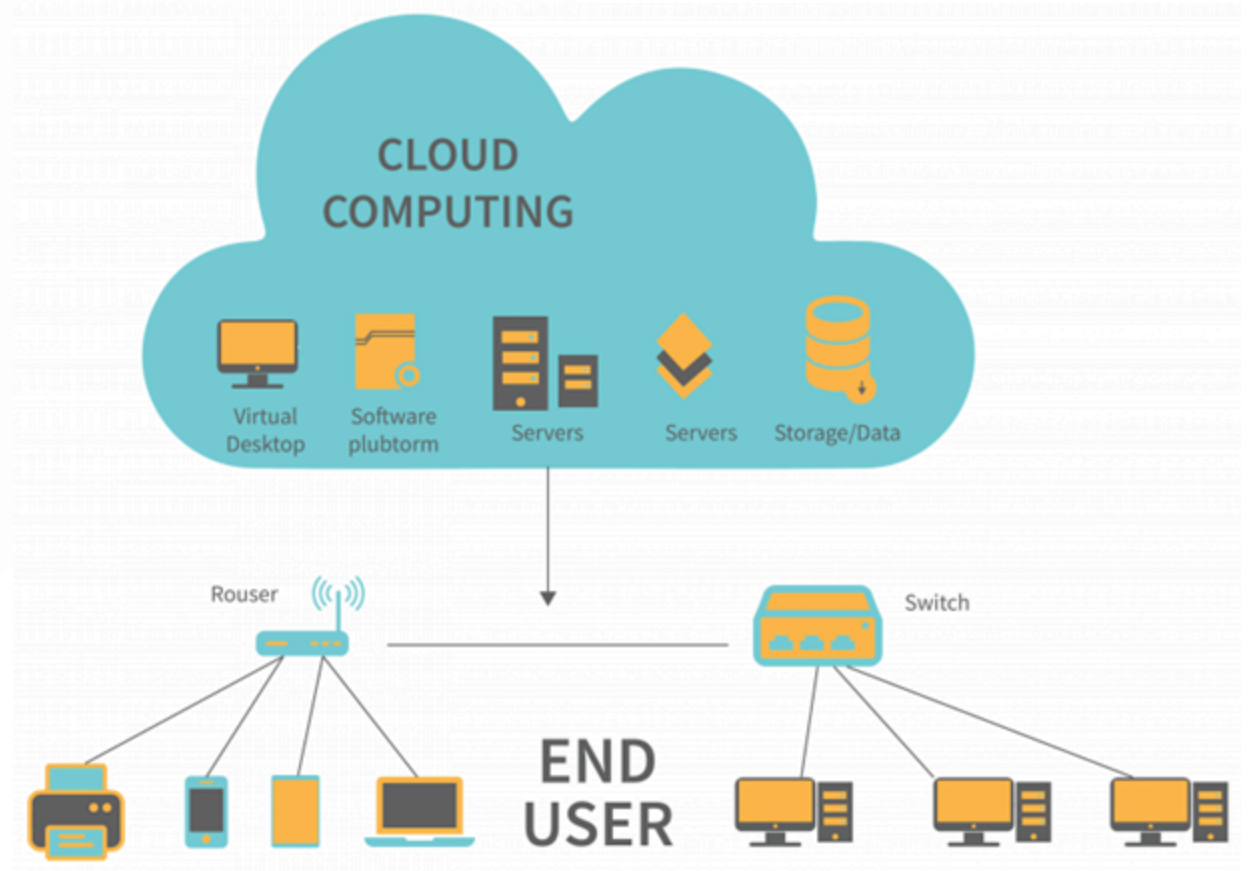
- Large amounts of created data need to be stored
- Data [types by access frequency](#):
 - Hot data: ns to ms
 - Warm data: ms to s
 - Cool data: min to 24 h
 - Cold data: days to weeks
 - Frozen data: weeks to years



[Forecast of cumulative stored data by type until 2030](#)

Data storage

- Where is data stored?
 - Endpoints: on-device memory for phones, laptops, desktops
 - Edge: institutional servers, offices
 - Core: data servers, cloud data centers
- Centralized data storage: AWS, Microsoft Azure, Google Cloud, Alibaba

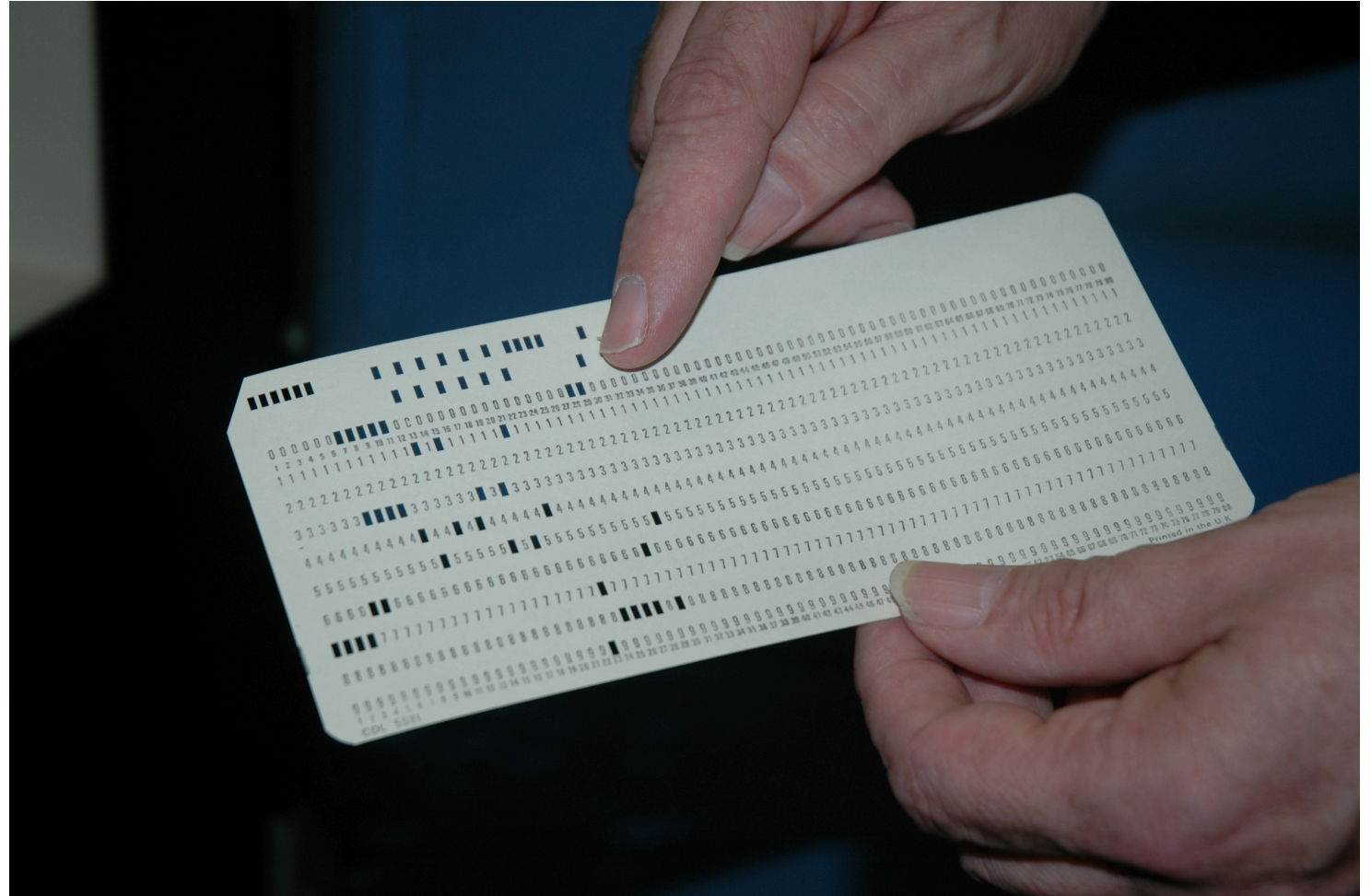


All cloud computing solutions include data storage

How do data storage technologies work?

Technologies: history

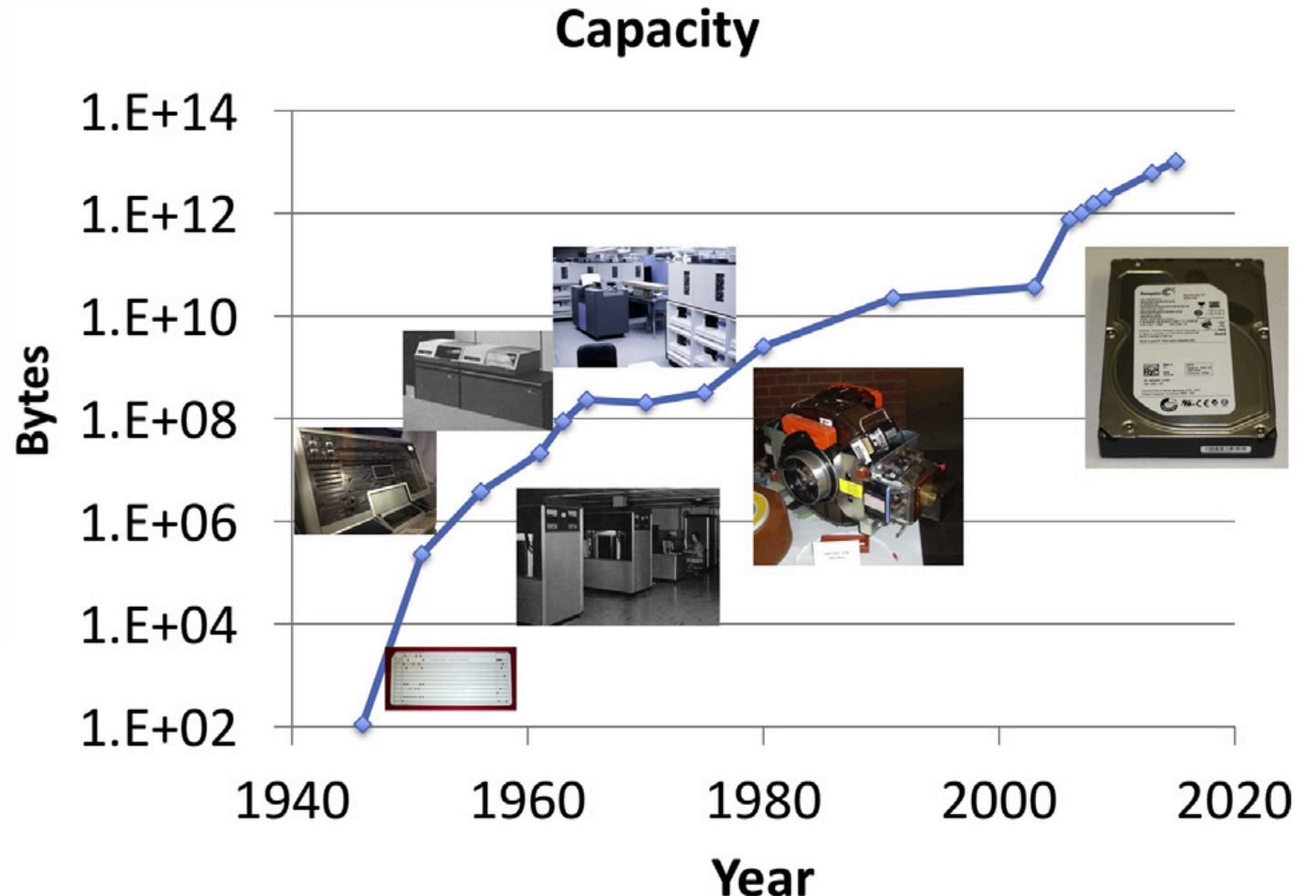
- [Punch cards](#): punched holes as on/off switches
- [Vacuum tubes](#): cathode-ray tubes as on/off switches
- Low bandwidth and storage capacity



[Punch cards were historically used to store data](#)

Technologies: modern options

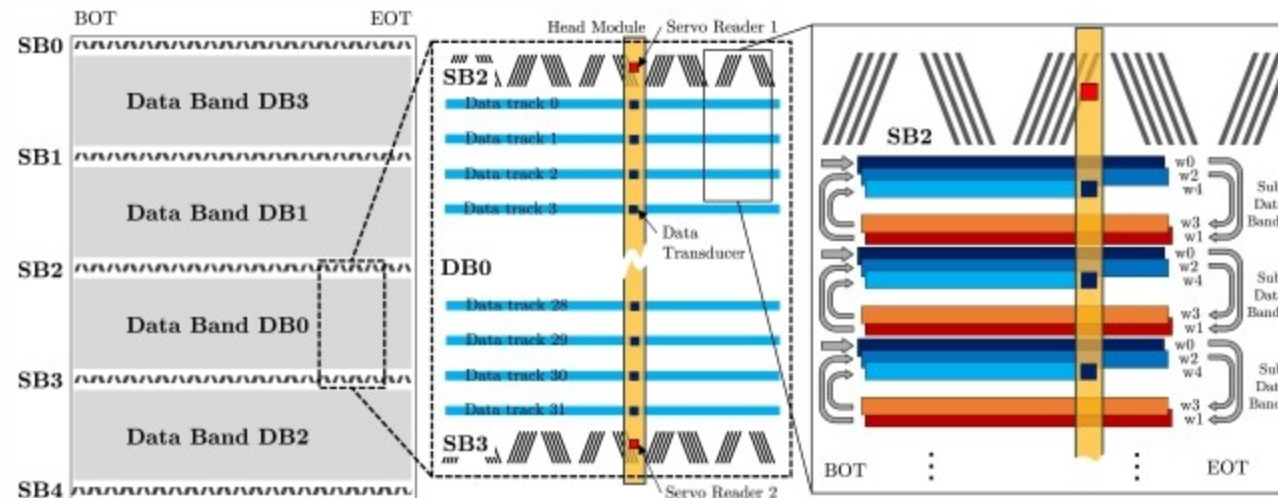
- Continuous [increase in data storage bandwidth and capacity](#):
 - Magnetic tape
 - Optical disc
 - Hard drive disk (HDD)
 - Flash memory (e.g. SSD)



[Growth in storage capacity from punch cards to SSDs](#)

Technologies: magnetic tape

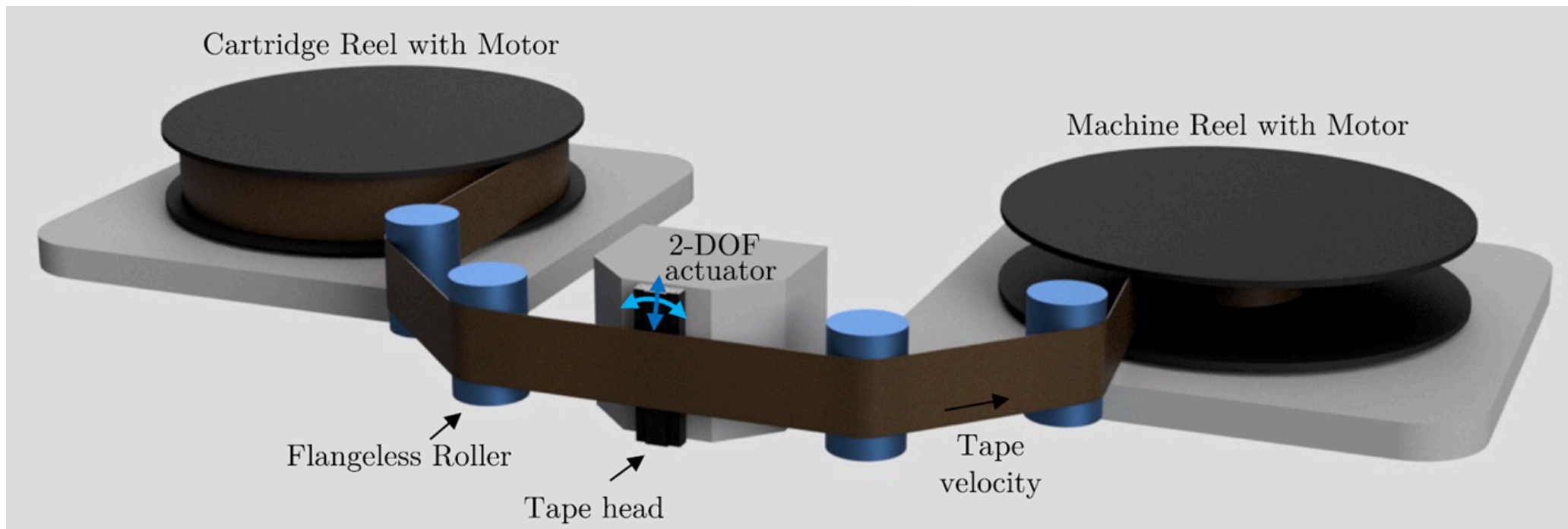
- Data stored in [magnetization patterns](#) down tape length
- Transducer detects magnetic field between regions with different magnetization
- Voltage pulses used to reconstruct information
- Format: linear tape open (LTO), latest generation with four data bands in parallel ([LTO-10](#))



[Magnetization patterns](#) are used to store data in magnetic tape

Technologies: magnetic tape

- Plastic substrate (PET, PEN, aramid) wound on tape reel, stored inside [cartridge](#)
- Robot arm grabs cartridge from archive, places into drive
- Drive spools up tape reel, tape head reads encoded data



[Reading data from a cartridge inside a tape drive](#)

Technologies: magnetic tape

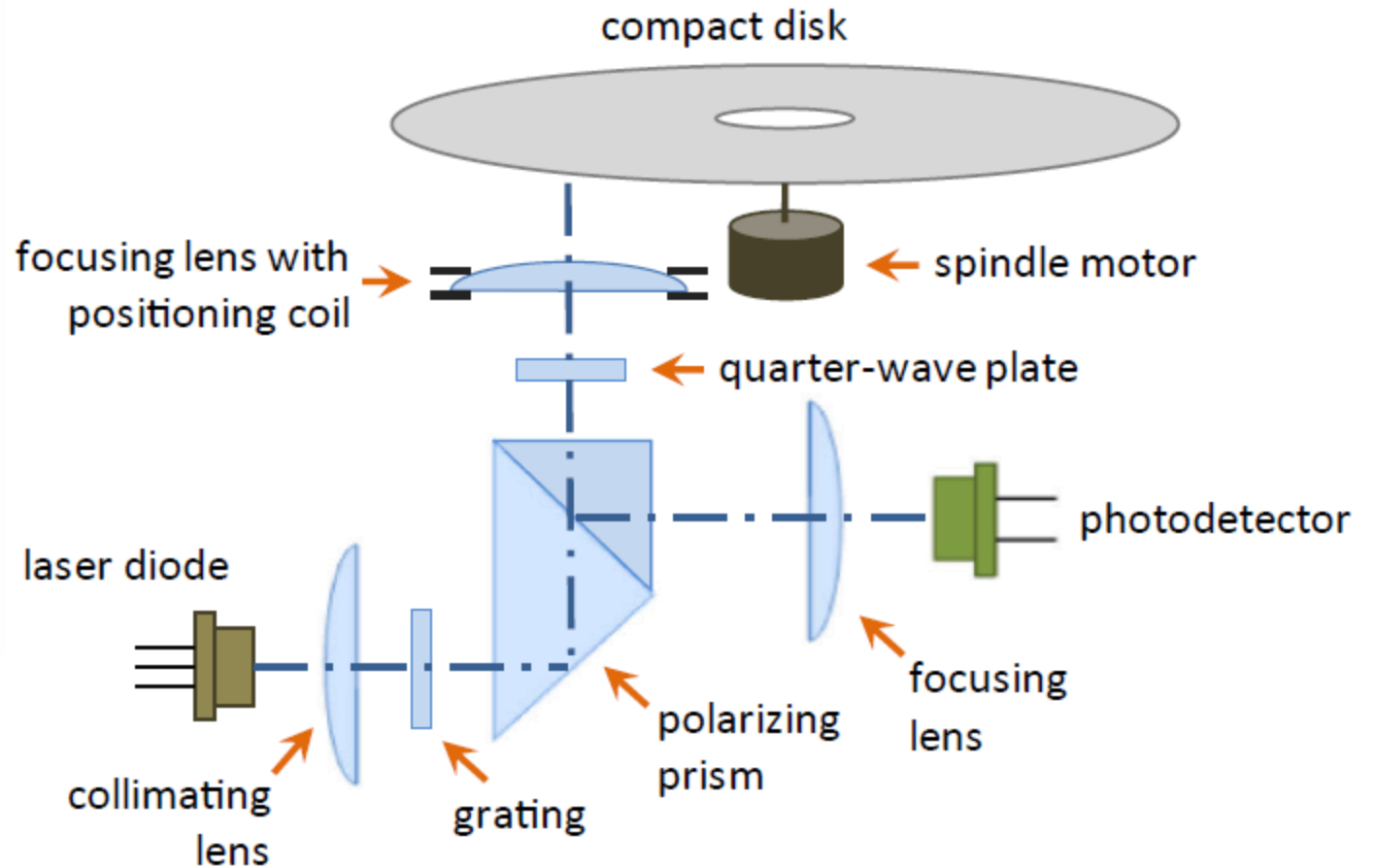
- Historically dominant in write-once, read-many (WORM) archive storage:
 - High volume density
 - Minimum [energy consumption](#) during standby (only for cooling)
 - High data rate
 - BUT: only [sequential reading](#), very high latency (need to extract cartridge physically!)



[Robot arm used in archival tape storage](#)

Technologies: optical discs

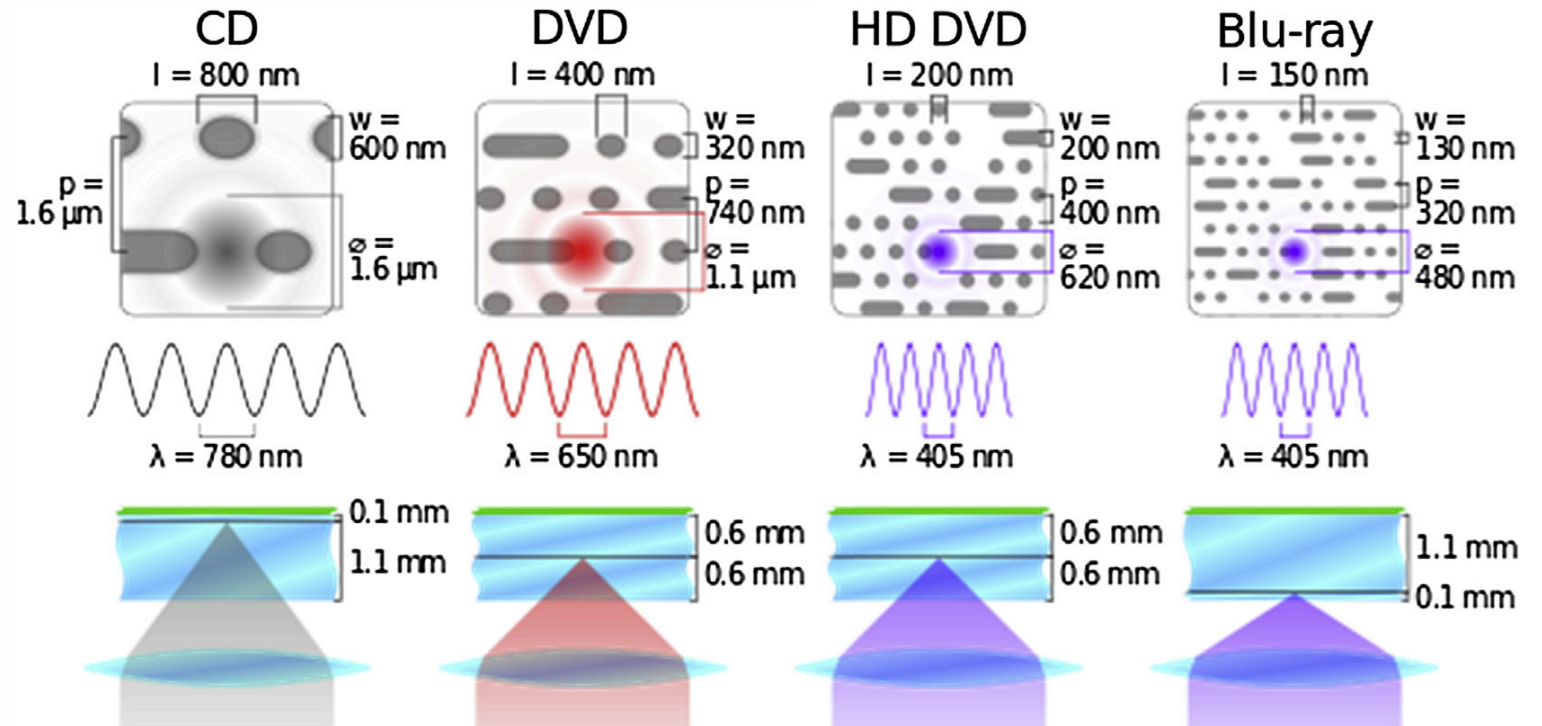
- CD: 1.2mm thick plastic disc with 120mm diameter
- Spiral pattern of elongated grooves [encodes data](#), covered with reflective metal layer
- Infrared laser beam to recover information from spinning disc



[Data is read off the grooves on the spinning disc](#)

Technologies: optical discs

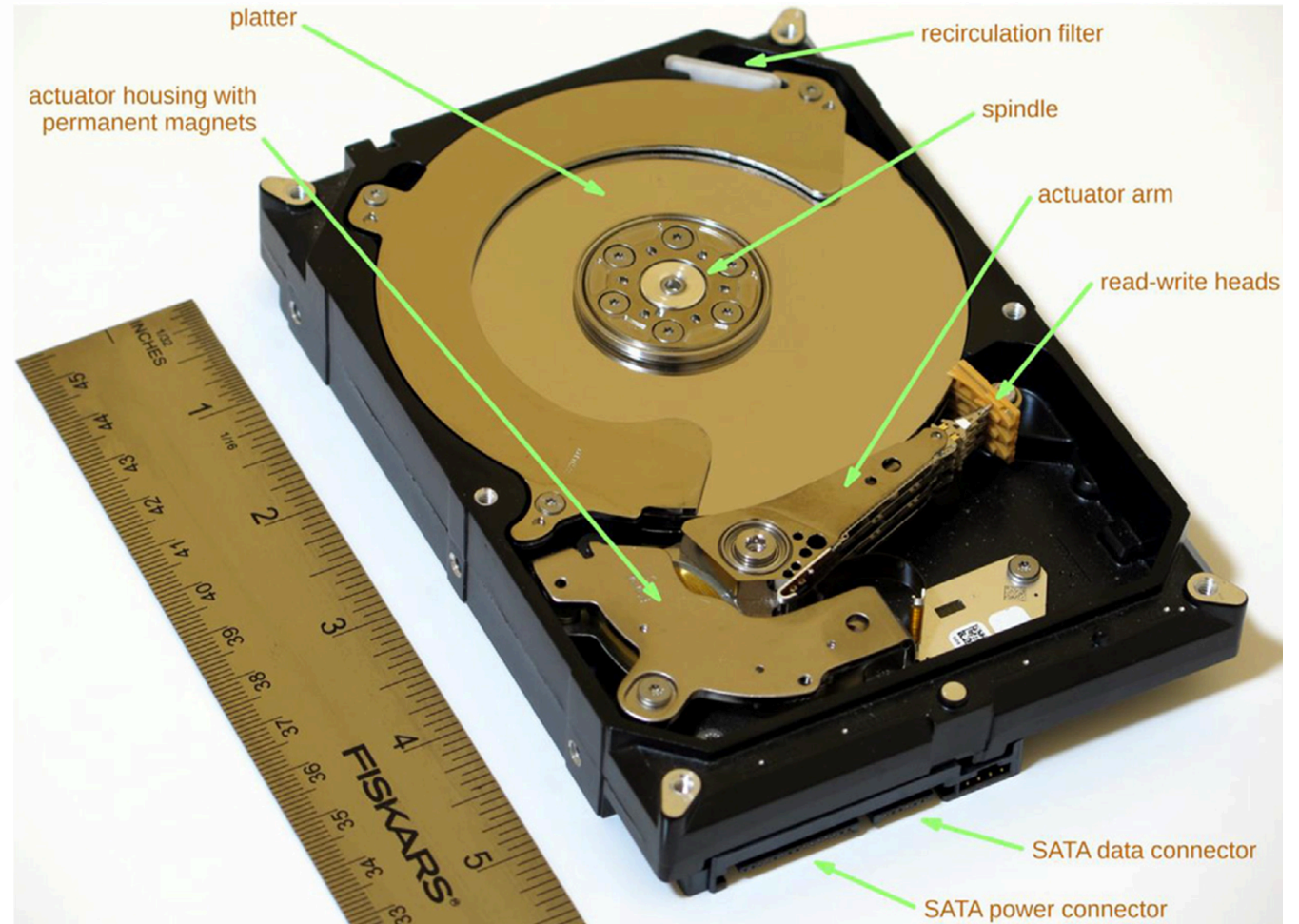
- CD-ROM: fixed content
- CD-R: recordable, can be 'burned' once
- CD-RW: rewritable, phase-change medium as reflective layer
- DVD, Blu-ray: increased data capacity ([up to 128 GB for quad-layer BDXL](#))



[Increasing data encoding density from CD to Blu-ray](#)

Technologies: HDDs

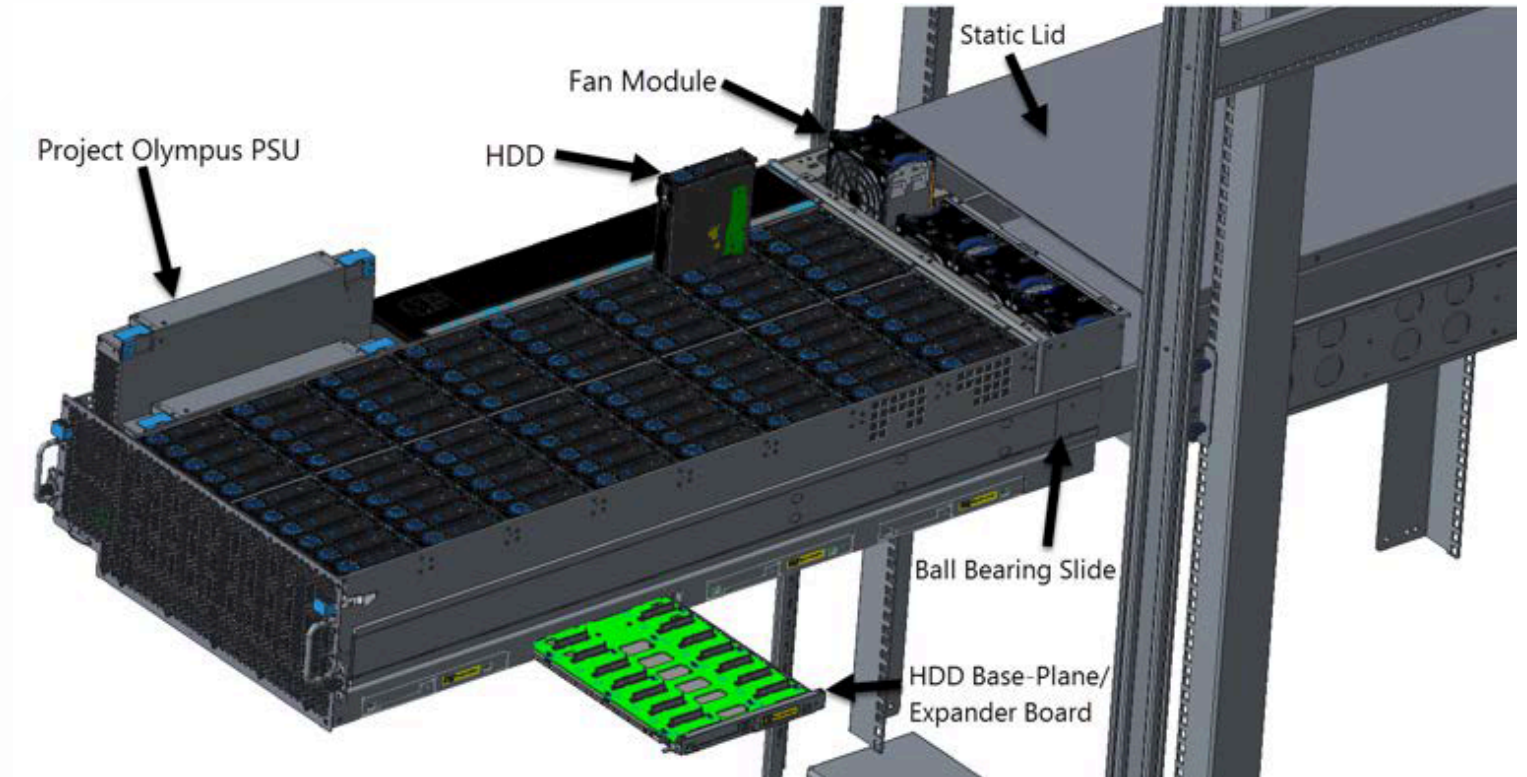
- Platter: disc with thin layer of permanent magnets
- Data stored as regions with different magnetization
- Multiple platters [spin together at several thousand RPM](#)
- Actuator arm for writing (changing magnetization of layer) or reading (outputting voltage pulse)



[HDD platters spin continuously to facilitate data access](#)

Technologies: HDDs

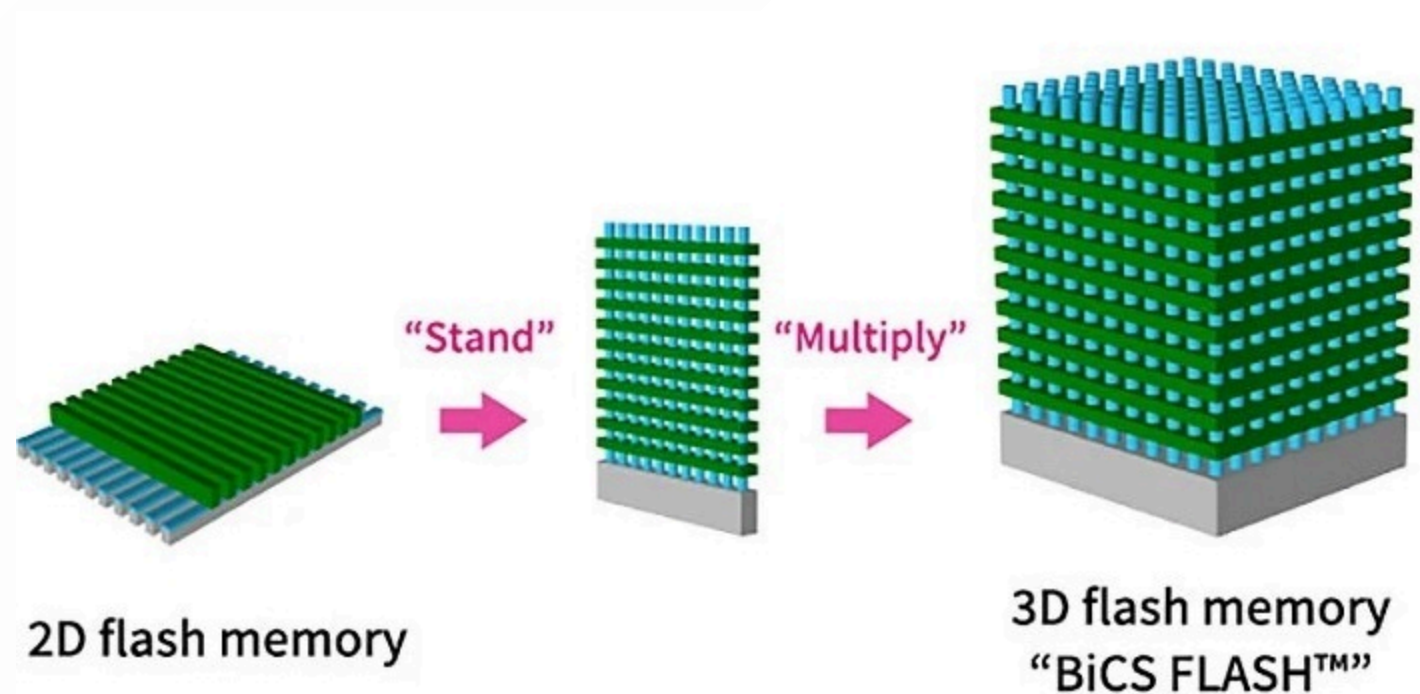
- High areal density: [above 800 Gbit per square inch](#)
- Moderate bandwidth: above 250 MB/s
- Moderate latency: depends on RPM, seek time
- Highest volume density of all storage technologies
- High energy consumption: [needs to keep spinning](#) to access data!



[Microsoft Azure 1.2 PB storage rack with 88 HDDs](#)

Technologies: flash memory

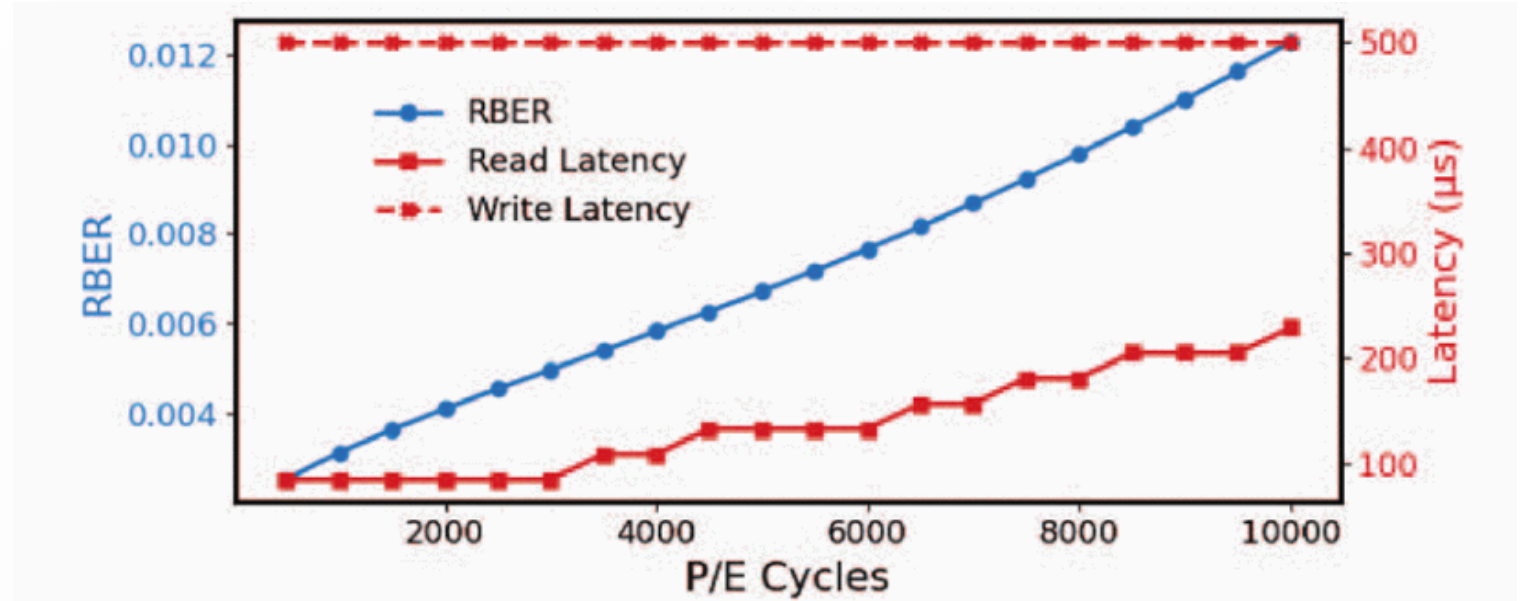
- Semiconductor memory: data stored as [electronic charge](#) in "memory cells" with transistors
- Threshold voltage of transistor increases when receiving charge
- SSDs: [stacking](#) of transistor arrays
- Reading: programmed cell does not conduct (0), empty cell conducts (1)



[Increased capacity can be obtained by stacking individual memory cells in multiple dimensions](#)

Technologies: flash memory

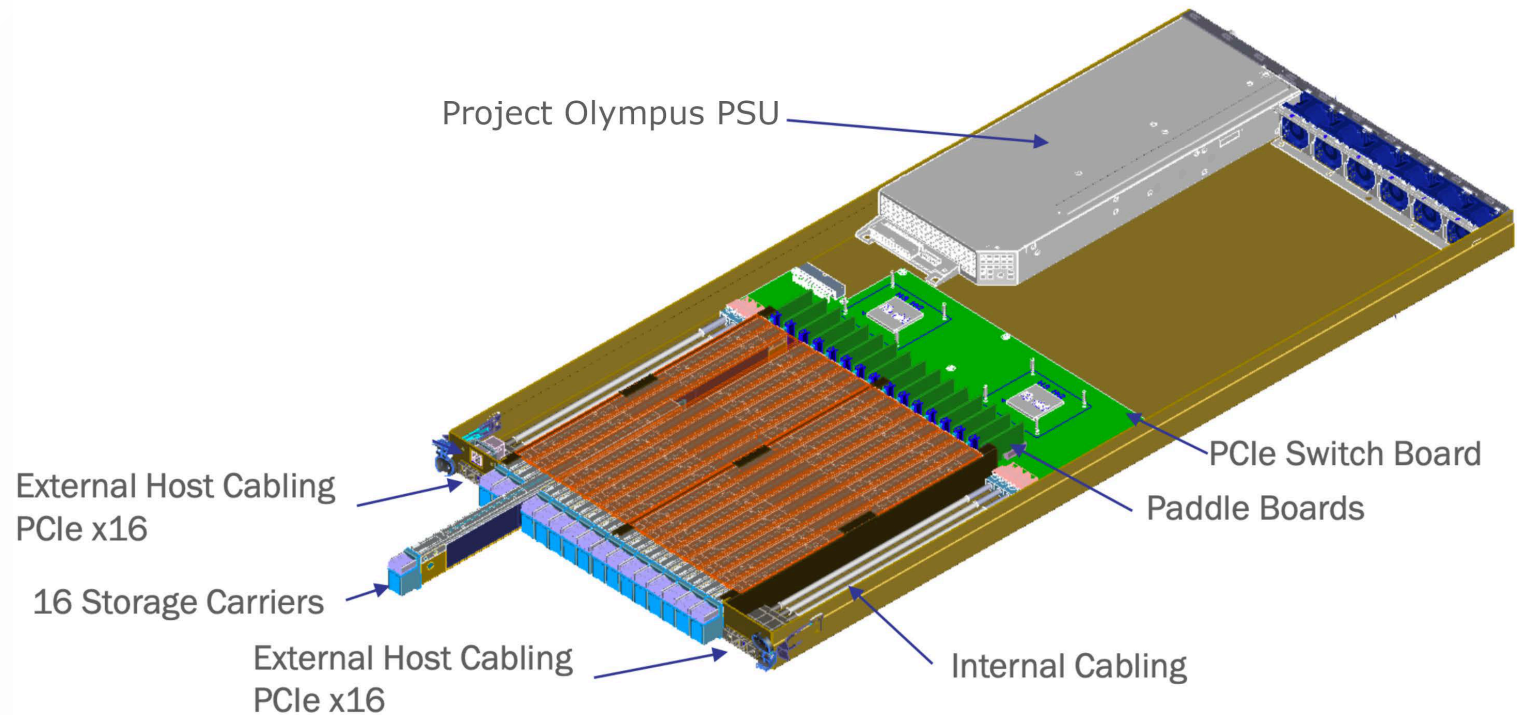
- Highly susceptible to defects → useful lifetime limited by number of read/write operations
 - Increasing latency with number of operations
- Typical metric: total volume of data handled over lifetime, terabytes written (TBW)



Raw bit error rate (RBER) and latency increase with each program/erase cycle

Technologies: flash memory

- Controller [required for multi-chip operations](#):
 - Slow operations by individual chips
 - Need logic, data buffer to distribute workloads
 - Need to distribute degradation between different chips
- No moving parts → higher IO operations per second (IOPS) than HDDs → [higher bandwidth!](#)

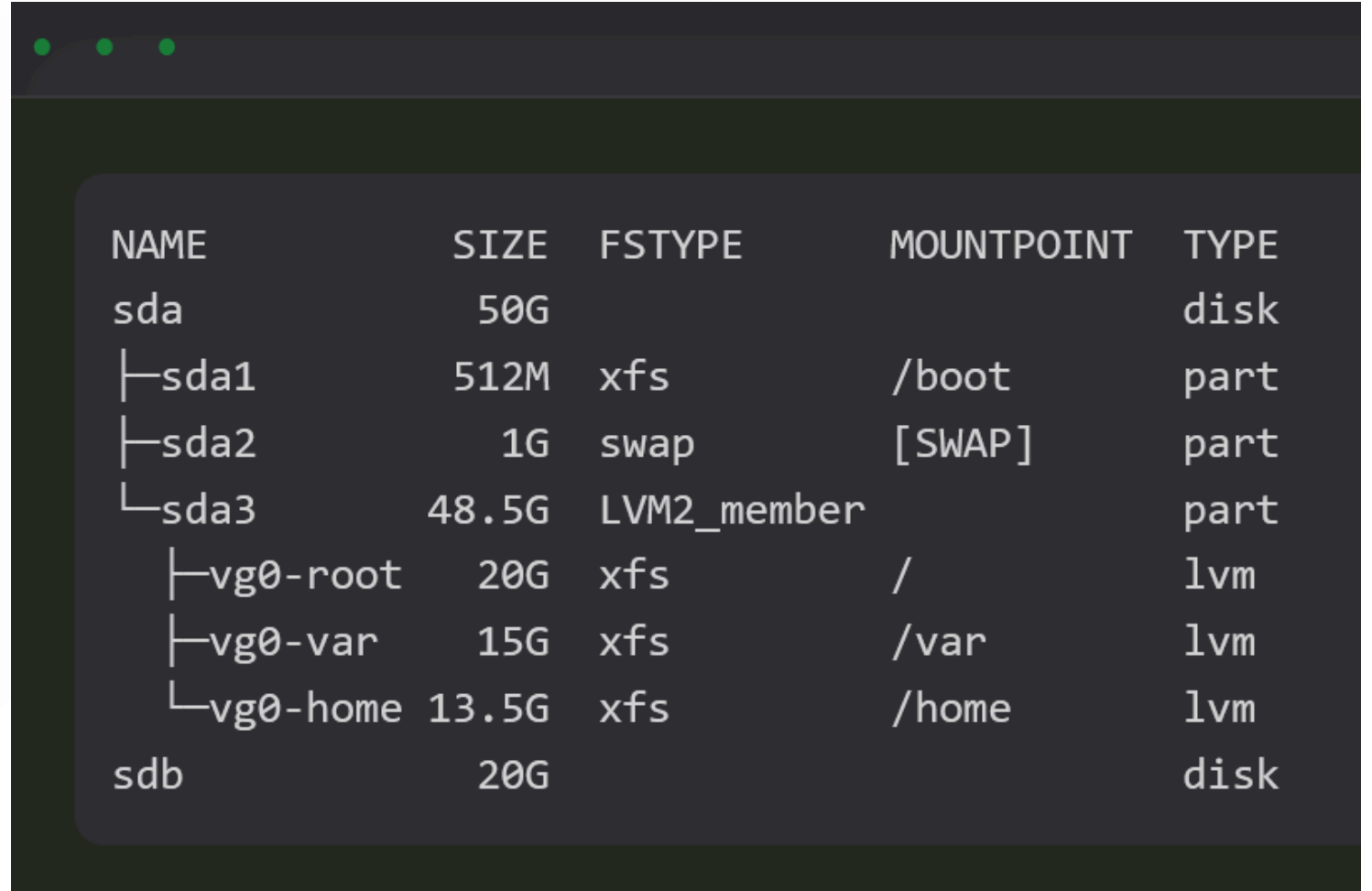


[Microsoft Azure 128 TB storage rack with 16 SSDs](#)

How do data storage architectures work?

Architectures: on-device data storage

- [Block devices](#) (disks): parallel representation of physical storage devices
- [Partitions](#): dividing a disk and applying a filesystem
- [Filesystem](#): architecture for structuring data, controlling read/write operations
- User only sees file tree!



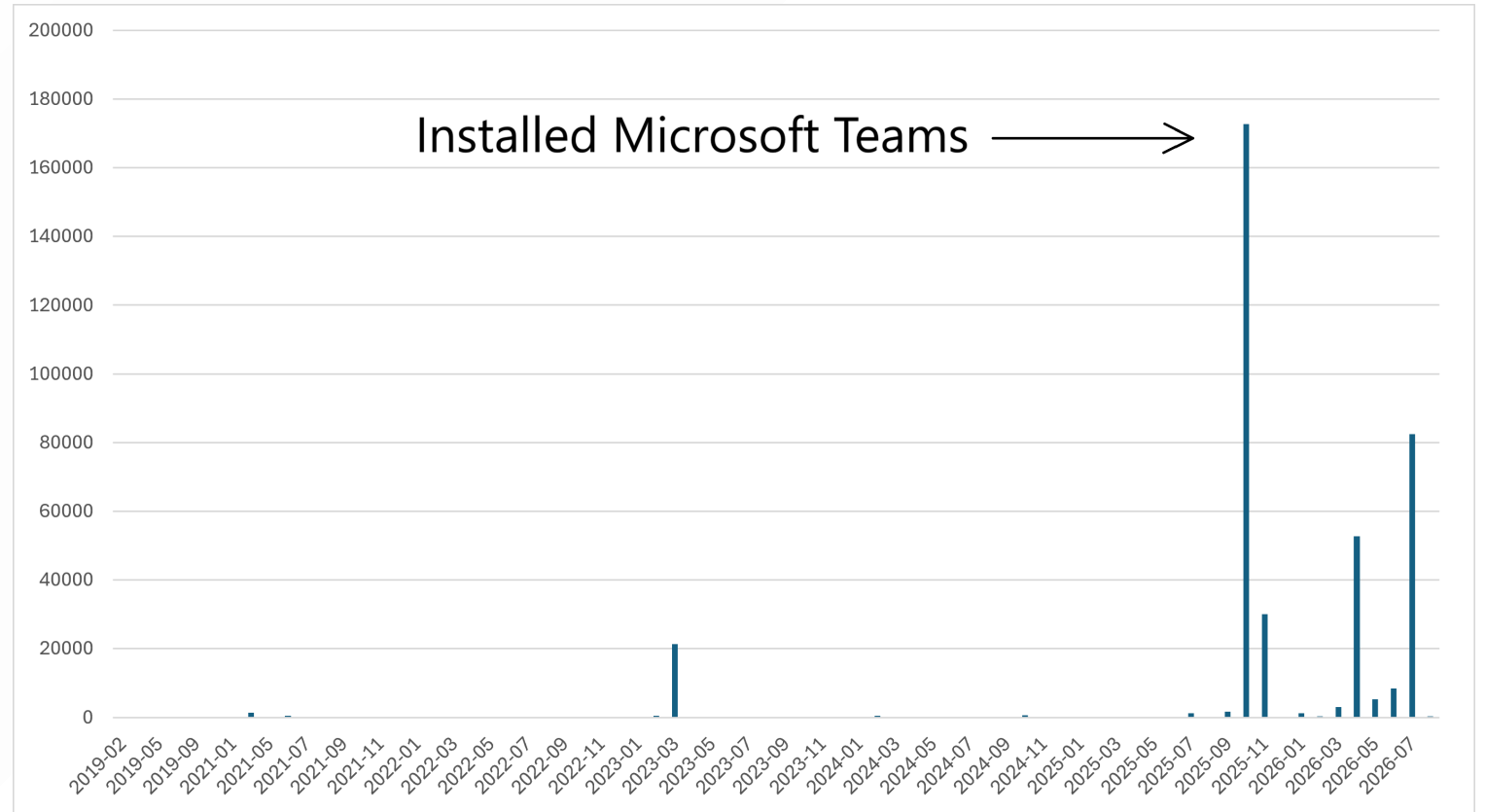
NAME	SIZE	FSTYPE	MOUNTPOINT	TYPE
sda	50G			disk
├─sda1	512M	xf	/boot	part
├─sda2	1G	swap	[SWAP]	part
└─sda3	48.5G	LVM2_member		part
├─vg0-root	20G	xf	/	lvm
├─vg0-var	15G	xf	/var	lvm
└─vg0-home	13.5G	xf	/home	lvm
sdb	20G			disk

[Visualizing storage disks and partitions with lsblk](#)

Exercise: data on your own device

How old are the files on your device?

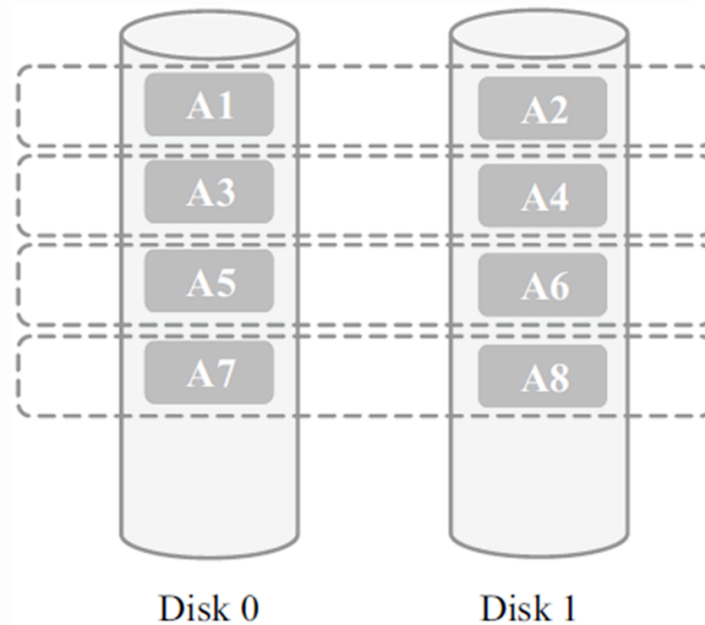
- Set up a PowerShell/terminal script that shows when files were created, then group by month



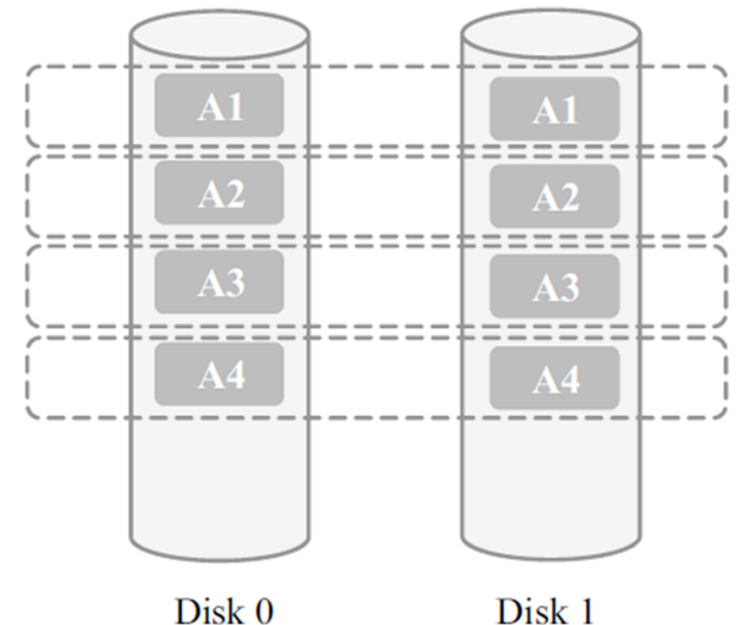
Example: monthly distribution of files created/modified

Architectures: RAID

- Redundant array of independent disks (RAID): [unifying multiple devices into single unit](#)
- RAID levels:
 - RAID 0: splitting files across disks
 - RAID 1: mirroring data to two disks
 - RAID 2-10: various clever ways of splitting files to ensure recovery
- Above RAID 0: more devices required to offer same available capacity



RAID 0



RAID 1

[RAID 1 requires twice the number of devices as RAID 0](#)

Implementing RAID on your own machine

- [mdadm](#) (Linux): combining multiple disk drives/partitions into a single device + filesystem
- Various RAID levels can be applied depending on desired redundancy

Output

/dev/md0:

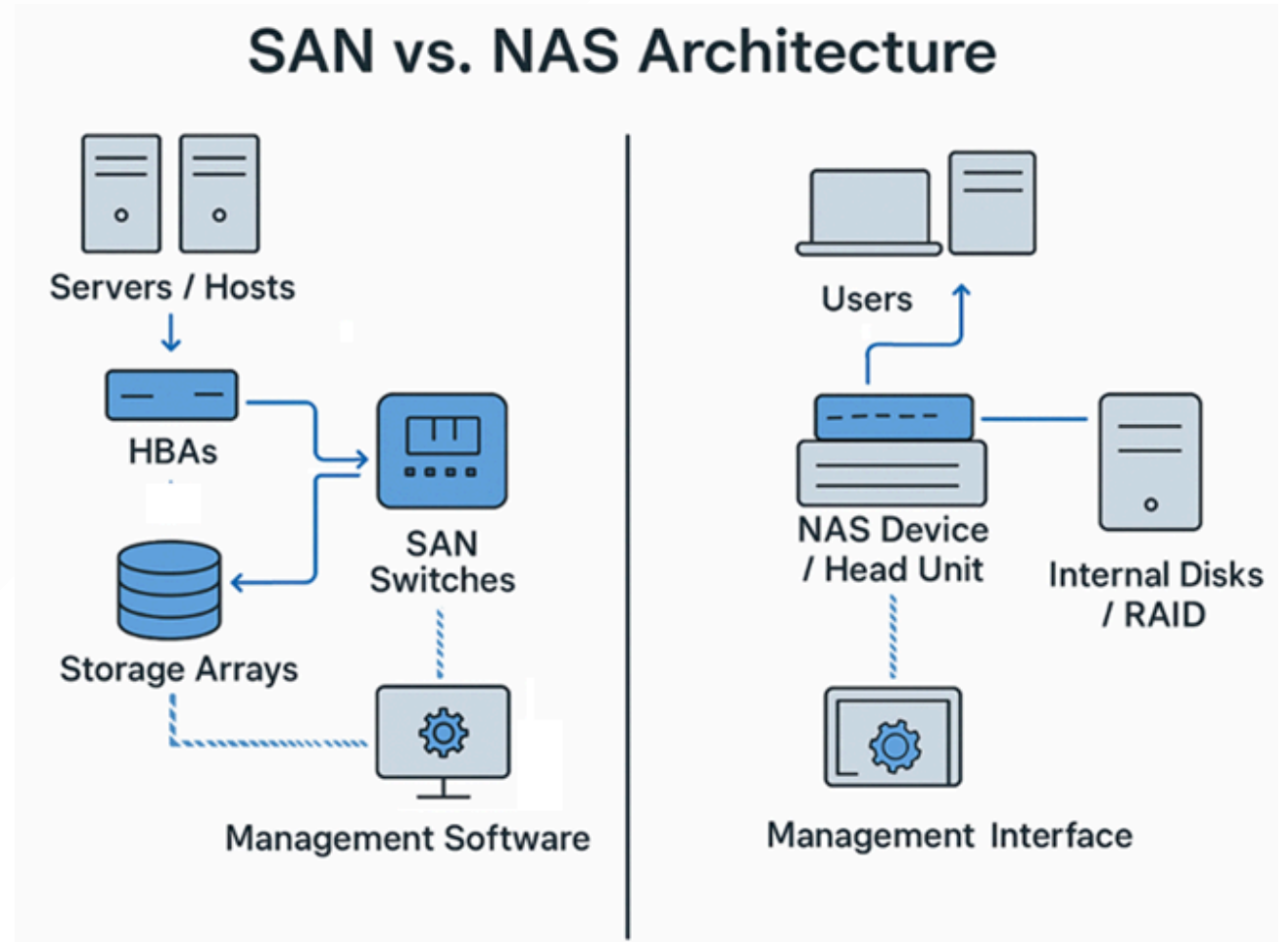
```
Version : 1.2
Creation Time : Thu Sep 29 17:07:10 2022
Raid Level : raid10
Array Size : 209582080 (199.87 GiB 214.61 GB)
Used Dev Size : 104791040 (99.94 GiB 107.31 GB)
Raid Devices : 4
Total Devices : 4
Persistence : Superblock is persistent
```

Number	Major	Minor	RaidDevice	State	
0	8	0	0	active sync set-A	/dev/sda
1	8	16	1	active sync set-B	/dev/sdb
2	8	32	2	active sync set-A	/dev/sdc
3	8	48	3	active sync set-B	/dev/sdd

[Visualizing a RAID10 array with four devices using `sudo mdadm -D /dev/md0`](#)

Architectures: software-hardware integration

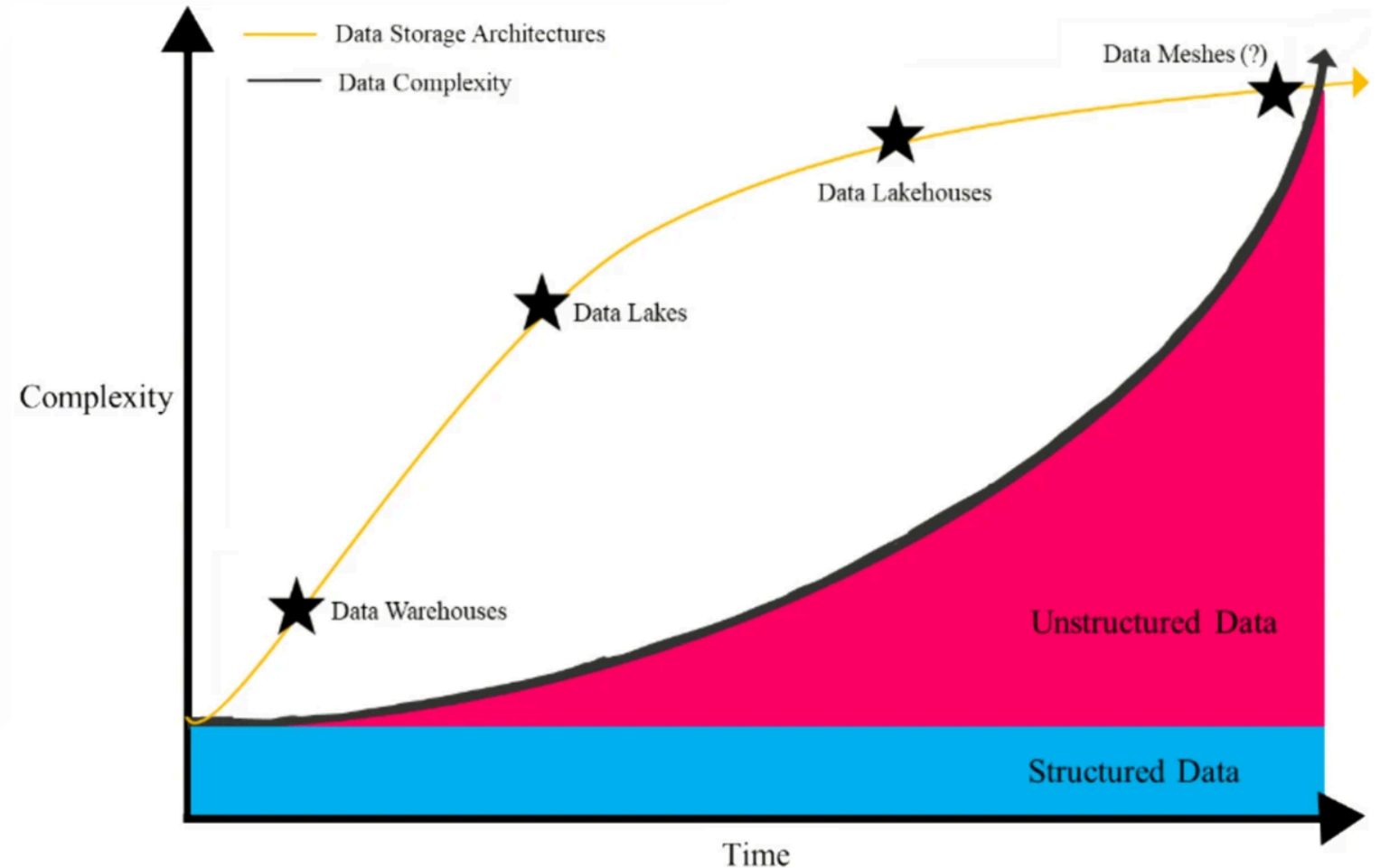
- [Abstractions](#) between users and storage hardware:
 - Device-attached storage (DAS): per-computer storage
 - Network-attached storage (NAS): file-level storage server connected to network
 - Storage area network (SAN): block-level network of dedicated storage devices
- Also require [networking](#) (Ethernet/Fibre Channel), file access systems



[Differences between the two main abstraction types](#)

Architectures: data management

- [Data management platform \(DMP\)](#): generations of storage architectures
 - Data warehouse: structured data in unified repository
 - Data lake: flexible repository for unstructured data
 - Data lakehouse: [storage layer](#) with open formats (Parquet), computing layer for structuring



[Management architectures evolve with data availability](#)

Recap

- More and more data is being created worldwide, leading to growing need for storage
- Data storage technologies (magnetic tape, HDDs, SSDs) have unique advantages and disadvantages driven by their operating principles
- Data storage architectures bridge the gap between hardware and user experience

Exam(ple) questions

- 1. An AI training workload includes hourly checkpoints, each with a 128 TB payload. If the workload takes 48 hours to complete and all checkpoints are stored in a RAID 1 configuration, how much storage capacity is required?**
 - a. 256 TB
 - b. 3072 TB
 - c. 1536 TB
 - d. 6144 TB
- 2. Assume that 150 zettabytes of data was created in 2025. If 1% of data created each year is stored, and the growth rate for data creation is 20%, how much capacity will be required to store the data created in 2050?**
 - a. 37.5 ZB
 - b. 143.1 ZB
 - c. 95.4 ZB
 - d. 1.5 ZB

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 - ~~d. 1.5 ZB~~

Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 7.1 slides/notes:



Thank you for your attention!

Next lecture: Sustainable Data Storage, Part 2

Friday, October 16, 15:45-17:30, EEMCS Hall F

- Environmental impacts: carbon footprint, HDD/SSD trade-offs, critical raw materials
- Industry trends: storage demand, areal density
- Prospective solutions: lifetime extension, data storage strategy, novel technologies

Questions?